Defect Report

BUG-0226 — Data Mismatch in Bluetooth Module: DataIntegrityError: field 'checksum...

1. Defect Summary

Defect ID	BUG-0226
Title	Data Mismatch in Bluetooth Module: DataIntegrityError: field 'checksum' expected...
Severity	Critical
Priority	Critical
Status	CLOSED
Component	Bluetooth Module
Application	APP-011 — ConnectedDevices
Reported Date	2024-02-21
Resolved Date	2023-09-28
Assigned To	ENG-032 — Helen Vargas

2. Traceability

Release	REL-005 — 5.1.9
Test Run	RUN-012 — Regression Cycle 12 — Build 7.0.4
Requirement	REQ-0234 — Data Synchronization — Health Metric Upload
Req Priority	Critical
Test Case ID	TC-0612
Test Case Title	Ensure that data synchronization does not exceed the p99 latency (1s) ...
Execution ID	EXEC-01750
Failure ID	FAIL-0358
Failure Type	Data Mismatch
Environment	ENV-001 — Mobile Lab
Build Version	6.0.4

3. Problem Description

During execution EXEC-01750, test case TC-0612 failed with Data Mismatch. The system exhibited data mismatch when the input data contains Unicode special characters.

4. Reproduction Steps

1. Record response time using the built-in performance monitor.
2. Re-connect the hub and confirm auto-recovery.
3. Simulate network drop using the network emulator profile '3G-Low'.
4. Re-connect the hub and confirm auto-recovery.
5. Assert that database record for entity 'ENT-307' has status='FAILED'.

SYNTHETIC DATA — No real Samsung confidential information

5. Expected vs Actual Behaviour

Expected	The system should complete voice command recognition within the defined SLA.
Actual	The system exhibited data mismatch when the input data contains Unicode special characters.

6. Root Cause

Health sensor sampling rate exceeds Bluetooth LE characteristic notification limit of 250ms intervals.

7. Resolution

Applied upstream codec patch; added resolution cap guard in pre-processing pipeline.

8. Related Historical Failures

HIST-0001	Data Mismatch	5.2.0	2021-12-08	13	Resolved
HIST-0008	Data Mismatch	4.1.8	2023-08-07	4	Wont Fix
HIST-0009	Data Mismatch	6.4.4	2023-08-23	7	Open

9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

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